机械制造基础 II

周计明

西北工业大学

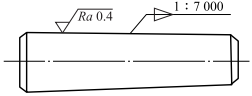
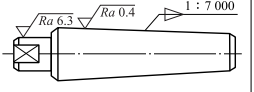
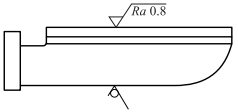
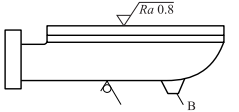
2026 年 · 秋

Chapter 9: 机械零件的结构工艺性

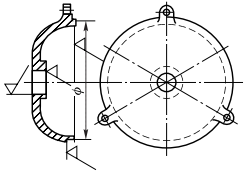
机械零件的结构工艺性

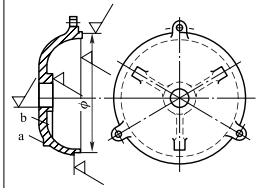
- 使用性能完全相同而结构不同的两个零件, 其加工方法与制造成本却可能有很大的差别
- **良好的结构工艺性** 是指根据使用要求所设计的零件结构, 能在同样的生产条件下, 采用 **高效率、低消耗和低成本** 的方法制造出来
- 满足零件的 **使用要求** 是考虑零件结构工艺性的前提，保证主要，照顾次要
  - 便于安装、加工
  - 减少切削加工量、降低成本
  - 减少加工辅助时间、减小工件变形
- 结构工艺具有相对性，如非圆孔对拉削来而言具有好的结构工艺性

便于安装、保证定位可靠

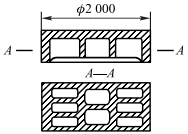
图例		说明
不合理的结构	合理的结构	
		锥度心轴一般是先车后磨，用顶尖、拨盘、卡箍装夹，应在心轴一端设计一圆柱表面，以便安装卡箍
		为了安装方便，增加了工艺凸台 B，精加工后，再把凸台 B 切除

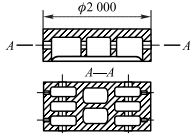
便于安装、保证定位可靠



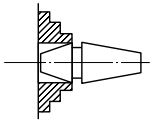


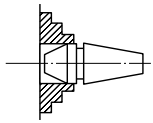
马达端盖无合适的装夹表面。一般在毛坯铸造时增设三个凸台 a，便于用三爪自定心卡盘装夹。设置筋板 b 为增加刚性，防止装夹时变形





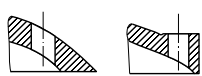
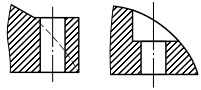
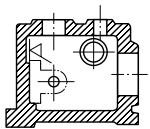
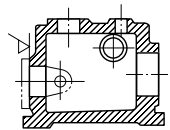
划线用大平板无合适装夹表面，在其两边各增加两个大孔，以便用压板螺栓压紧工件且便于吊装起运



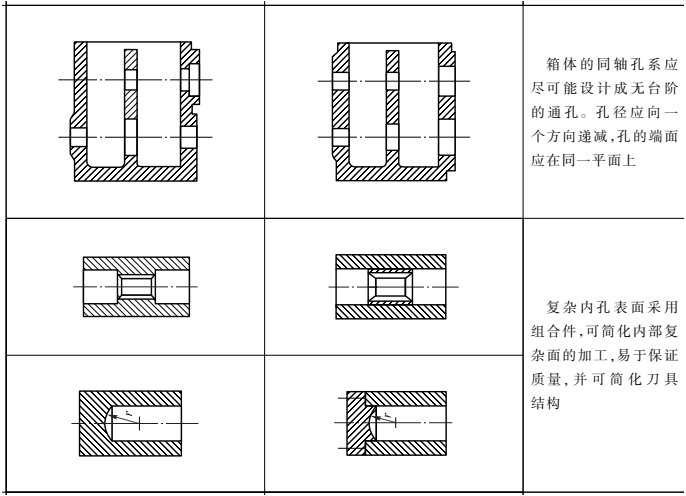


原设计用圆锥面作装夹部位，无法夹牢。改进设计后，以圆柱面装夹，装夹稳固，定位可靠

减小加工难度

		工件上钻头进出表面应与孔的轴线垂直，否则会加大钻孔难度，甚至可能折断钻头
		将箱体内表面的加工改为外表面的加工，可使加工大为方便

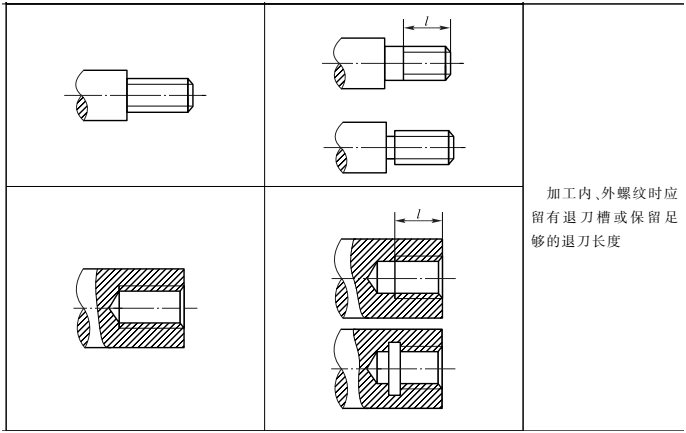
减小加工难度



箱体的同轴孔系应尽可能设计成无台阶的通孔。孔径应向一个方向递减,孔的端面应在同一平面上

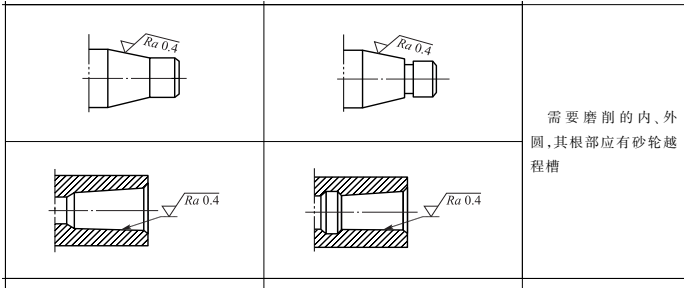
复杂内孔表面采用组合件,可简化内部复杂面的加工,易于保证质量,并可简化刀具结构

便于进刀和退刀



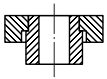
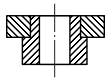
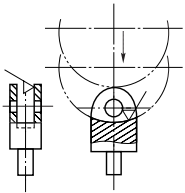
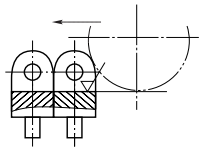
加工内、外螺纹时应留有退刀槽或保留足够的退刀长度

便于进刀和退刀

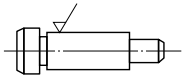
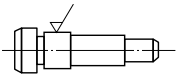
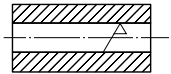
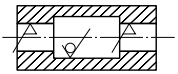


需要磨削的内、外圆，其根部应有砂轮越程槽

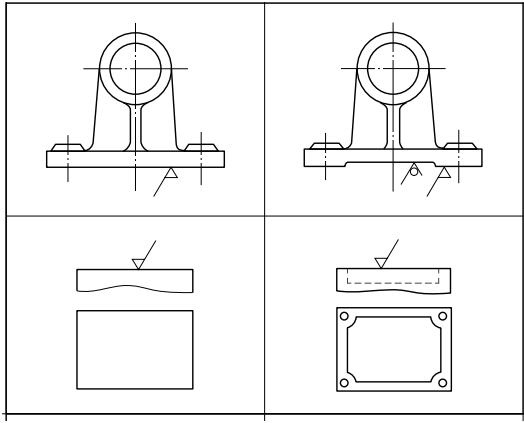
力求简单的几何形状

		把阶梯孔改成简单的孔，便于加工
		沟槽底部若是圆弧，铣刀直径必须与工件圆弧直径一致；槽底若为平面，则可选择任何直径的铣刀加工。槽底为平面，还可多件串联起来同时加工，提高生产率

减少加工表面面积

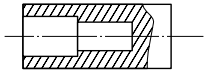
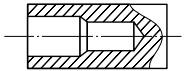
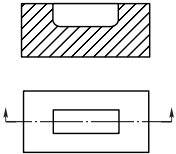
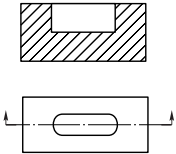
		当轴与盘套类零件相配合时,应保证配合部位的精度,非配合表面不必制成高精度,这样不仅可减少精加工面积,且易于保证质量
		当孔的长度与直径之比较大时,应保证与轴相配合部位的精度,以节省材料和降低工时

减少加工表面面积

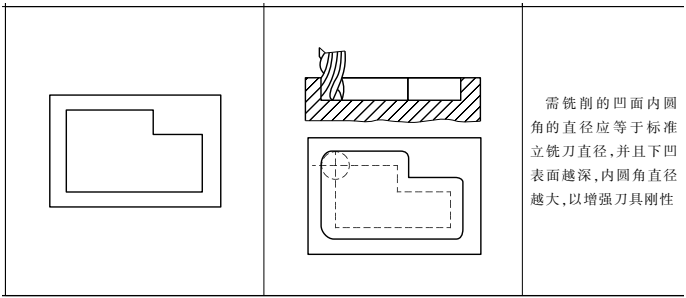


轴承座、箱体、机架等零件的底平面，应设计成中部成凹状的平面，以减少加工面积，并保证工作可靠

便于采用标准刀具

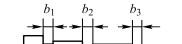
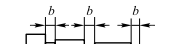
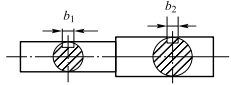
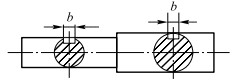
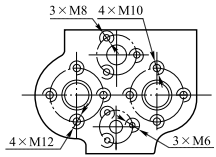
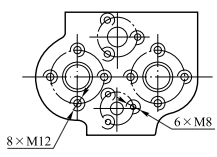
		孔应设计成标准直径，以便选用标准钻头等定径刀具；直径过渡应采用与钻头顶角相同的锥度
		槽的形状（直角、圆角）和尺寸应与立铣刀形状相符

便于采用标准刀具

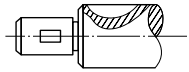
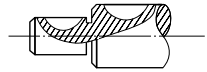
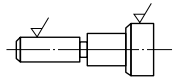
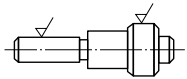
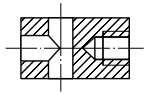
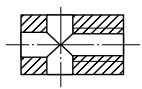
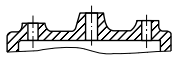
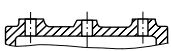


需铣削的凹面内圆角的直径应等于标准立铣刀直径，并且下凹表面越深，内圆角直径越大，以增强刀具刚性

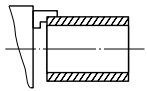
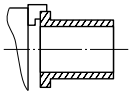
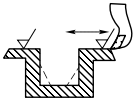
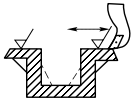
减少加工辅助时间

减少刀具种类和换刀次数			同一零件上结构相同的槽(键槽、刀槽),其宽度(包括内圆角半径)应尽可能一致,以减少刀具种类和换刀次数
			
			箱体上的螺纹孔直径应尽量一致或减少种类,以便采用同一丝锥或减少丝锥规格

减少加工辅助时间

减少操作时间			键槽的尺寸、方位应尽量一致，便于在一次走刀中铣出各键槽
减少安装次数			需要磨削或精车的零件，应考虑在一次安装中完成，以保证加工精度和节省工时
			原设计需要二次装夹，改进设计后一次装夹即可，且易于保证孔的同轴度
减少对刀次数			零件表面上的凸台，应尽可能布置在同一平面上，以便在一次对刀中加工出各凸台

使零件有足够的刚度

考虑夹紧力			薄壁、套筒类零件夹紧时易变形，若一端加凸缘，可提高工件刚度
考虑加工时冲击力			设置加强筋，可提高零件刚度，减少刨削或铣削时工件的振动或变形
	